

W

Depth and Optical Flow Methods Study

Lirui Wang, Computer Vision, CSE455 Spring 2018

Paul G. Allen School of Computer Science & Engineering

3D Scene Understanding

- Navigation
- Manipulation
- ...



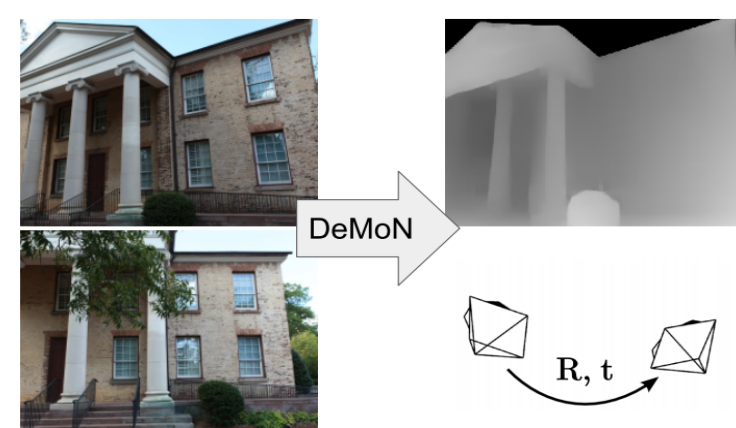
Scene Flow

- Depth
 - Distance
 - Structure
- Optical Flow
 - Motion

Research Methods

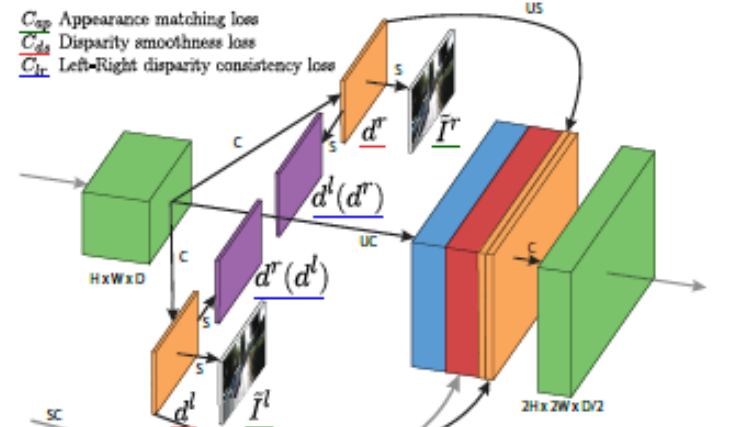
- ✓ CNN Predicts Depth
- ✓ Encoder-Decoder, Skip-Connection
- ✓ Smoothness Loss
- ✓ Adam Optimizer, BatchNorm, Low LR

DeMoN [1]



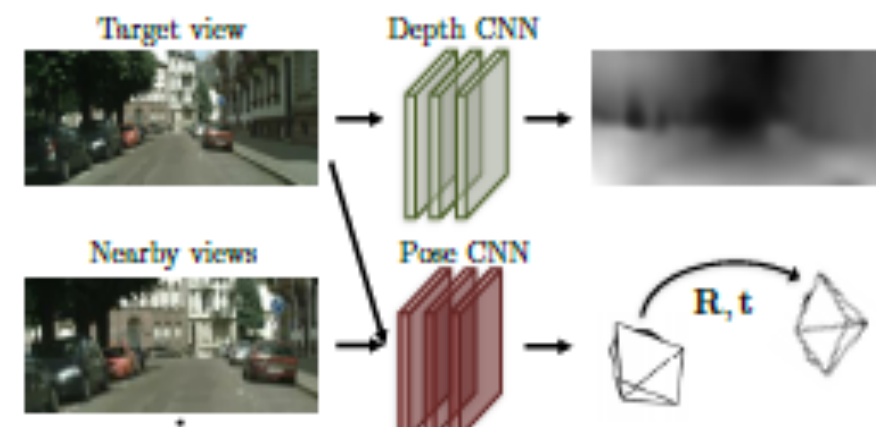
Supervised

MonoDepth [2]



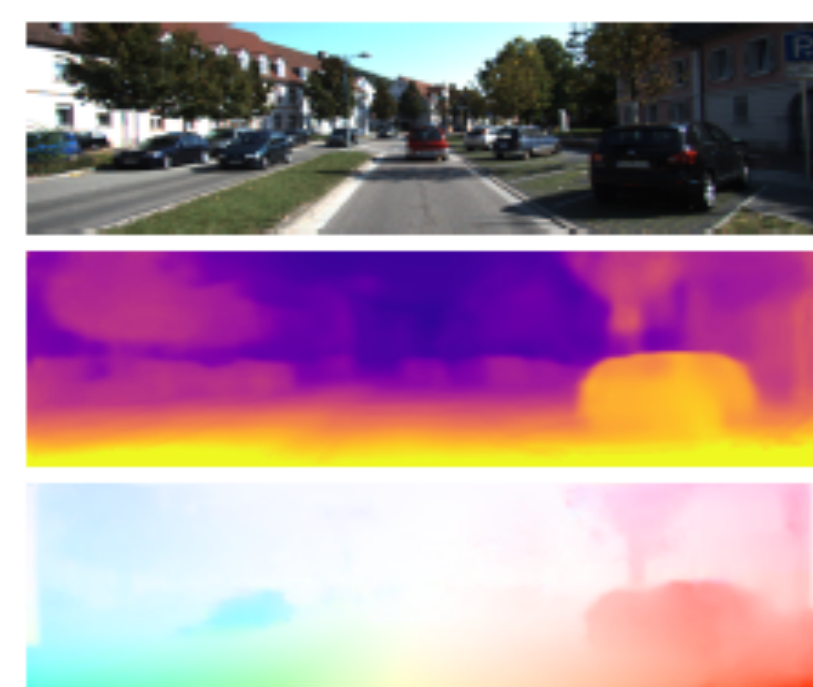
Stereo Camera

SfMLearner [3]

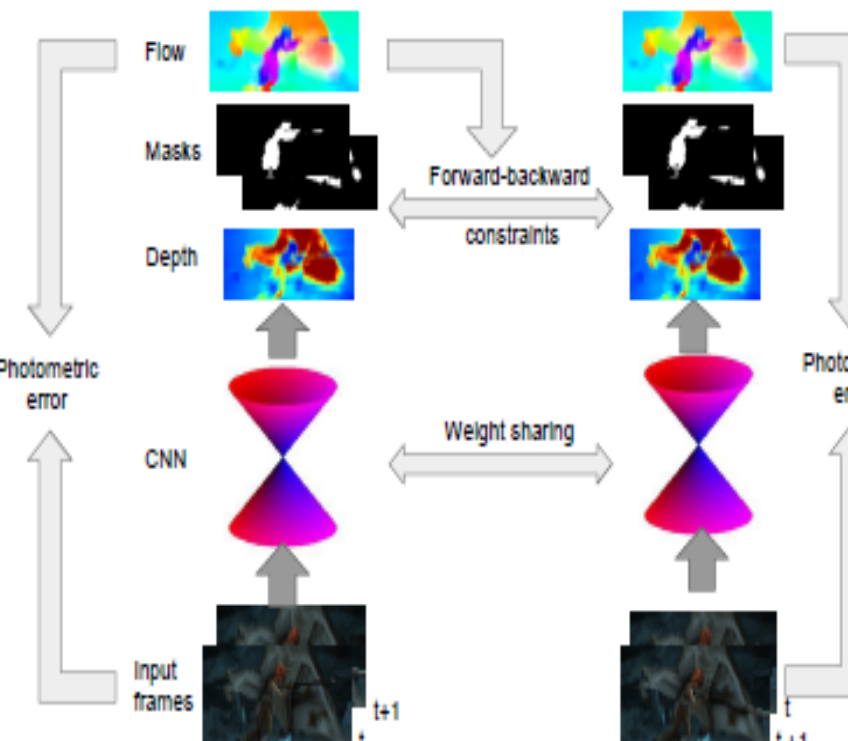


Unsupervised

GeoNet [4]



SfM-Net [5]



Various Supervision

- Alternates With Optical Flow
- Iterative Network Structure
- Scale-Invariant Gradient Loss
- Stage-wise Training Schedule
- Synthetic and Real Datasets

- Stereo Supervised Signal
- Left-Right Consistency
- Reconstruction Loss
- Multiple Output Scales
- Kitti & Make3D Datasets

- Unstructured Video Sequence
- Geometric Projection Idea
- Differentiable Warping Loss
- Explanability Confidence Mask
- Trained on Kitti & CityScape

- SfMLearner & FlowNet2 [5]
- Nature of 3D Scene Geometry
- Rigid Structure Reconstructor
- Non-rigid Motion Localizer
- Geometric Consistency Loss

- Depth & Per-Frame Point Cloud
- Scene Motion & Flow
- Forward Backward Constraints
- Objects Motion Masks
- Allow Depth, Pose & Optical Flow Supervision or Unsupervised

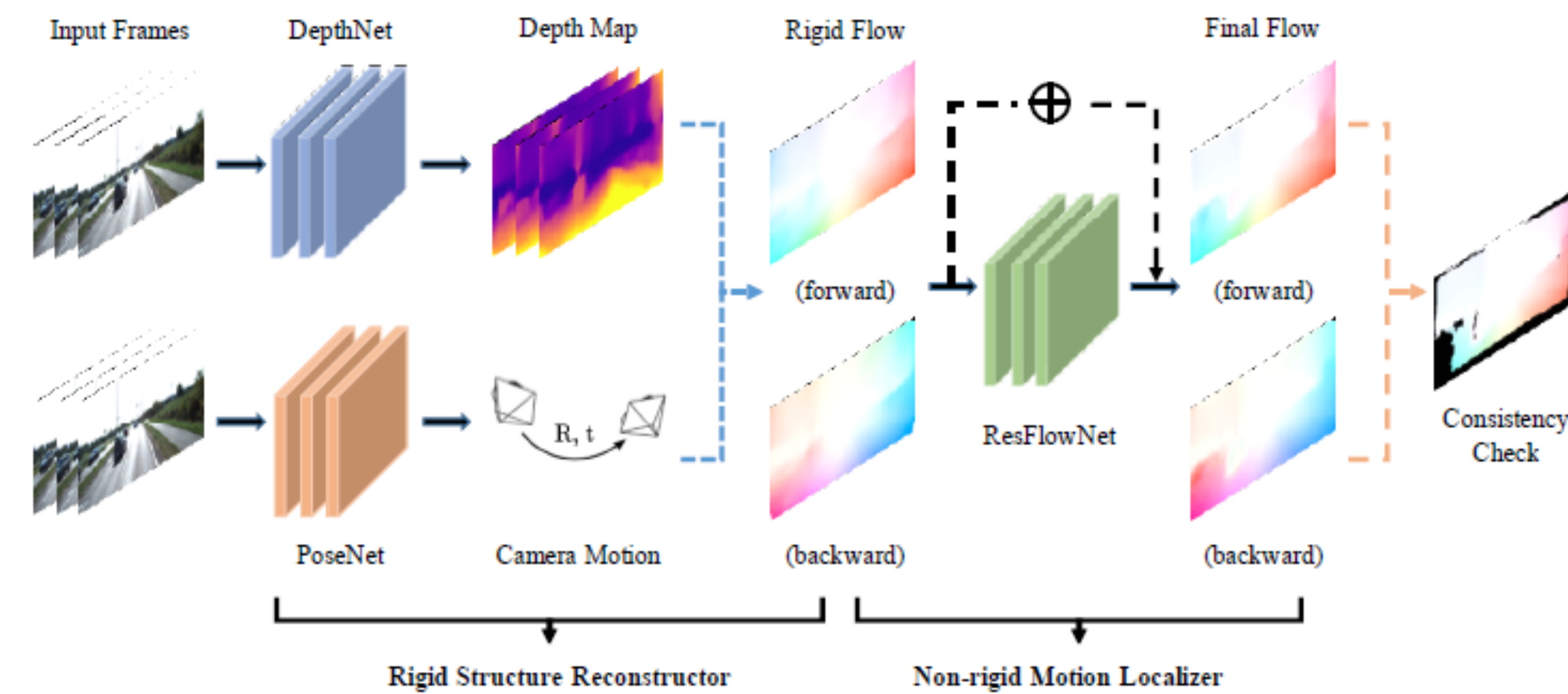
- ✗ Unscaled Depth
- ✓ Surface Normal
- ✓ Pose
- ✗ Optical Flow

- ✗ Unscaled Depth
- ✗ Surface Normal
- ✓ Pose
- ✗ Optical Flow

- ✗ Unscaled Depth
- ✗ Surface Normal
- ✓ Pose
- ✓ Optical Flow

- ✗ Unscaled Depth
- ✗ Surface Normal
- ✓ Pose
- ✓ Optical Flow
- ✓ Segmentation

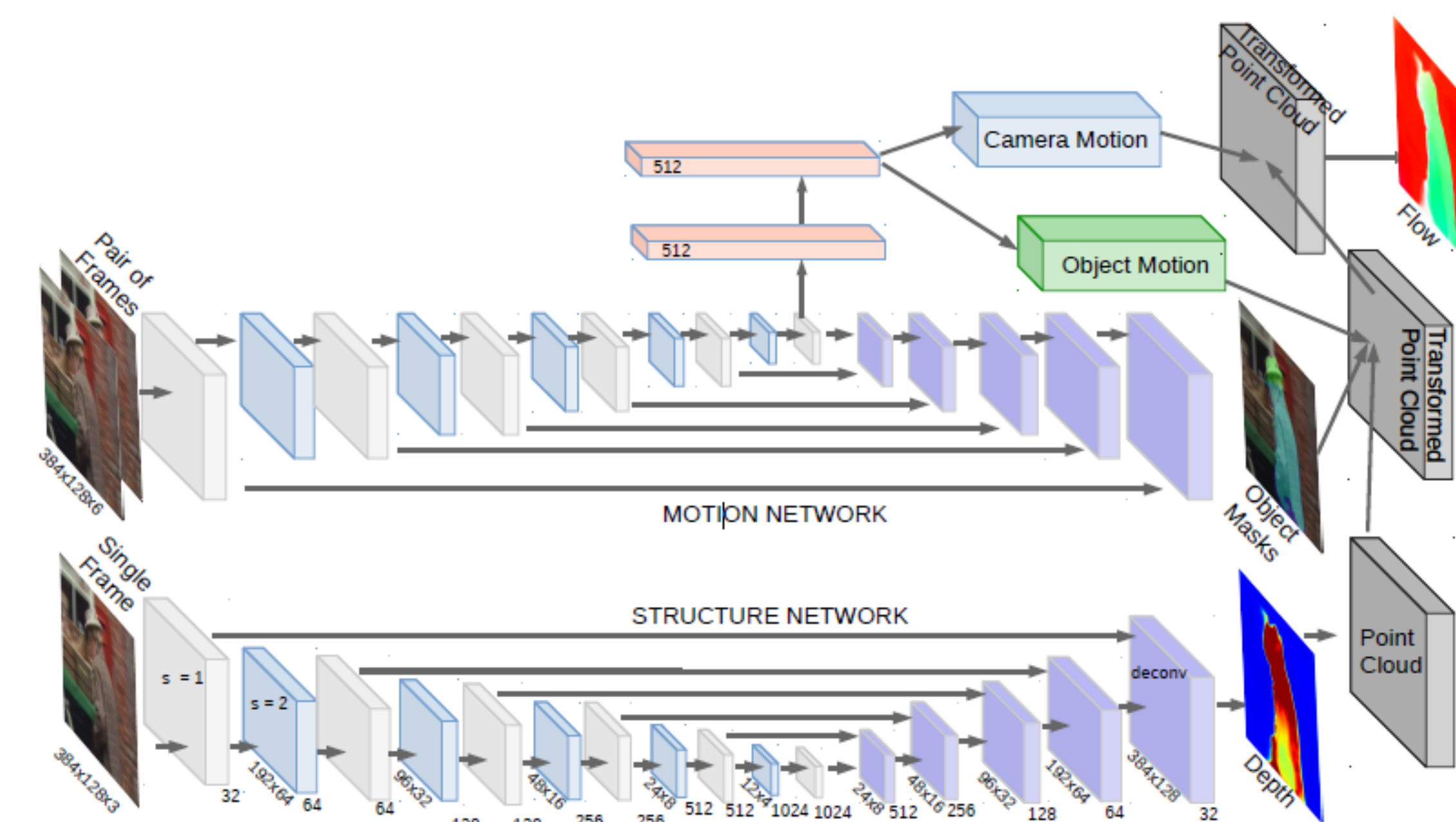
GeoNet Architecture



Method Insights

- Rigid Flow (PoseNet and DepthNet In SfMLearner)
- Residual Flow
- Photometric Loss $\mathcal{L}_{rw} = \alpha \frac{1 - SSIM(I_t, \hat{I}_t^{rig})}{2} + (1 - \alpha) \|I_t - \hat{I}_t^{rig}\|_1$
- Edge-Aware Smoothness Loss $\mathcal{L}_{ds} = \sum_{p_t} |\nabla D(p_t)| \cdot (e^{-\nabla I(p_t)})^T$
- Geometric Consistency Loss $\mathcal{L}_{gc} = \sum_{p_t} [\delta(p_t)] \cdot \|\Delta f_{t \rightarrow s}^u(p_t)\|_1$

SfM-Net Architecture



Method Insights

- Backproject to Point Cloud $\mathbf{X}_t^i = \begin{bmatrix} X_t^i \\ Y_t^i \\ Z_t^i \end{bmatrix} = \frac{d_t^i}{f} \begin{bmatrix} \frac{x_t^i}{d_t^i} - c_x \\ \frac{y_t^i}{d_t^i} - c_y \\ f \end{bmatrix}$
- Apply Object Transformation $\mathbf{X}_t^i = \bar{\mathbf{X}}_t + \sum_{k=1}^K \mathbf{m}_t^k(i) (R_t^k(\mathbf{X}_t - p_k) + t_t^k - \mathbf{X}_t)$
- Apply Camera Pose $\mathbf{X}_t^i = R_t^c(\mathbf{X}_t^i - p_t^c) + t_t^c$
- Project to Image Plane $\begin{bmatrix} \frac{x_{t+1}^i}{y_{t+1}^i} \\ \frac{w}{h} \end{bmatrix} = \frac{f}{Z_t^i} \begin{bmatrix} X_t^i \\ Y_t^i \\ f \end{bmatrix} + \begin{bmatrix} c_x \\ c_y \end{bmatrix}$
- Calculate Loss

Reference

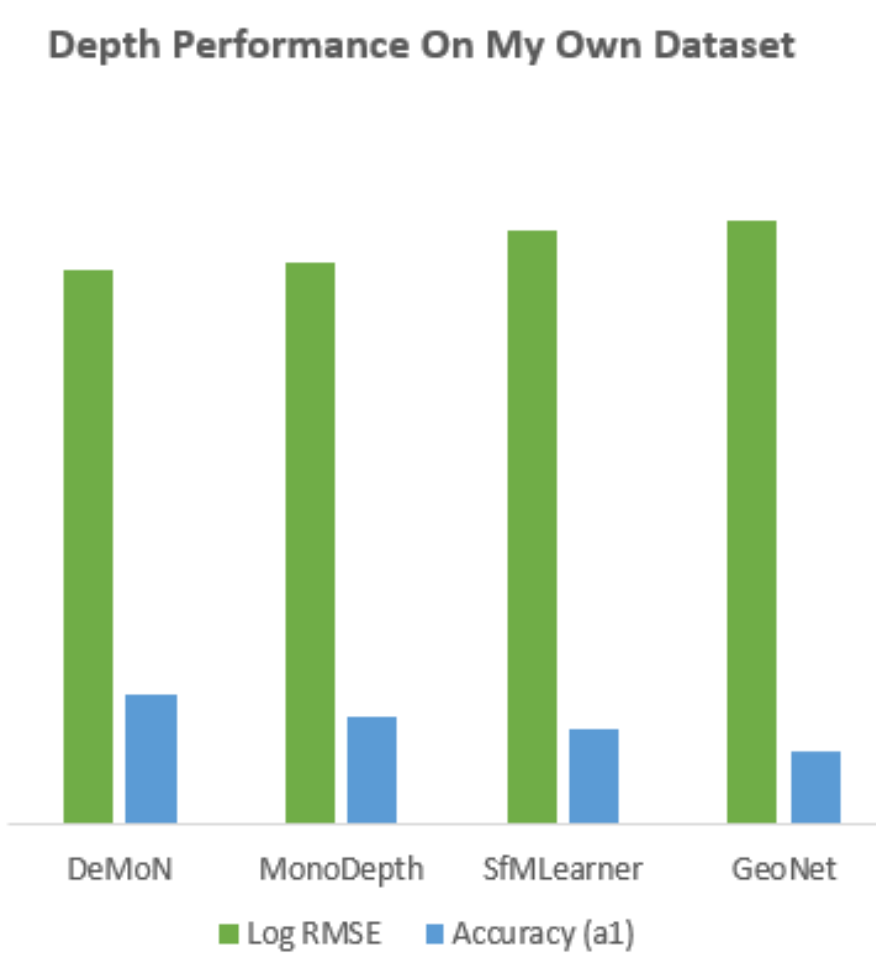
- [1] B. Ummenhofer, H. Zhou, J. Uhrig, N. Mayer, E. Ilg, A. Dosovitskiy, and T. Brox. Demon: Depth and motion network for learning monocular stereo. In CVPR, 2017.
- [2] C. Godard, O. Mac Aodha, and G. J. Brostow. Unsupervised monocular depth estimation with left-right consistency. In CVPR, 2017.
- [3] T. Zhou, M. Brown, N. Snavely, and D. G. Lowe. Unsupervised learning of depth and ego-motion from video. In CVPR, 2017.
- [4] Z. Yin and J. Shi. GeoNet: Unsupervised Learning of Dense Depth, Optical Flow and Camera Pose. In CVPR, 2018.
- [5] V. Sudheendra, R. Susanna, S. Cordelia, S. Rahul & F. Katerina. (2017). SfM-Net: Learning of Structure and Motion from Video.
- [6] E. Ilg, N. Mayer, T. Saikia, M. Keuper, A. Dosovitskiy, and T. Brox. FlowNet 2.0: Evolution of optical flow estimation with deep networks. In CVPR, 2017.
- [7] C. Wang, J. Buenaposada, R. Zhu, S. Lucey. (2017). Learning Depth from Monocular Videos using Direct Methods.

Experiments

Depth Prediction Evaluation Results [4]

Method	Supervised	Dataset	Abs Rel	Sq Rel	RMSE	RMSE log	$\delta < 1.25$	$\delta < 1.25^2$	$\delta < 1.25^3$
Godard et al. [15]	Pose	K	0.148	1.344	5.927	0.247	0.803	0.922	0.964
Zhou et al. [56]	No	K	0.208	1.768	6.856	0.283	0.678	0.885	0.957
Zhou et al. [56] updated ¹	No	K	0.183	1.595	6.709	0.270	0.734	0.902	0.959
Ours VGG	No	K	0.164	1.303	6.090	0.247	0.765	0.919	0.968
Ours ResNet	No	K	0.155	1.296	5.857	0.233	0.793	0.931	0.973

Taken From GeoNet's Paper. GeoNet Outperforms SfMLearner And Comparable To MonoDepth in Kitti. DeMoN has not been tested on Kitti. SfM-Net trained with stereo pairs is equivalent to the approach of MonoDepth..



However, Not Generalized At All.

Depth Prediction Comparisons in Indoor Scenes



- DeMoN Generalizes Slightly Better
- Indoor Scene, Static Camera Limits
- Unacceptable Results From Others
- Scale Ambiguity [7] of Mono-Depth

Analysis on Flow Prediction



- Decent Performance Extracts Objects
- Outliers Exist In The Test Sets
- GeoNet's Result Implies that Cascade FlowNet Can Correct Object Motion

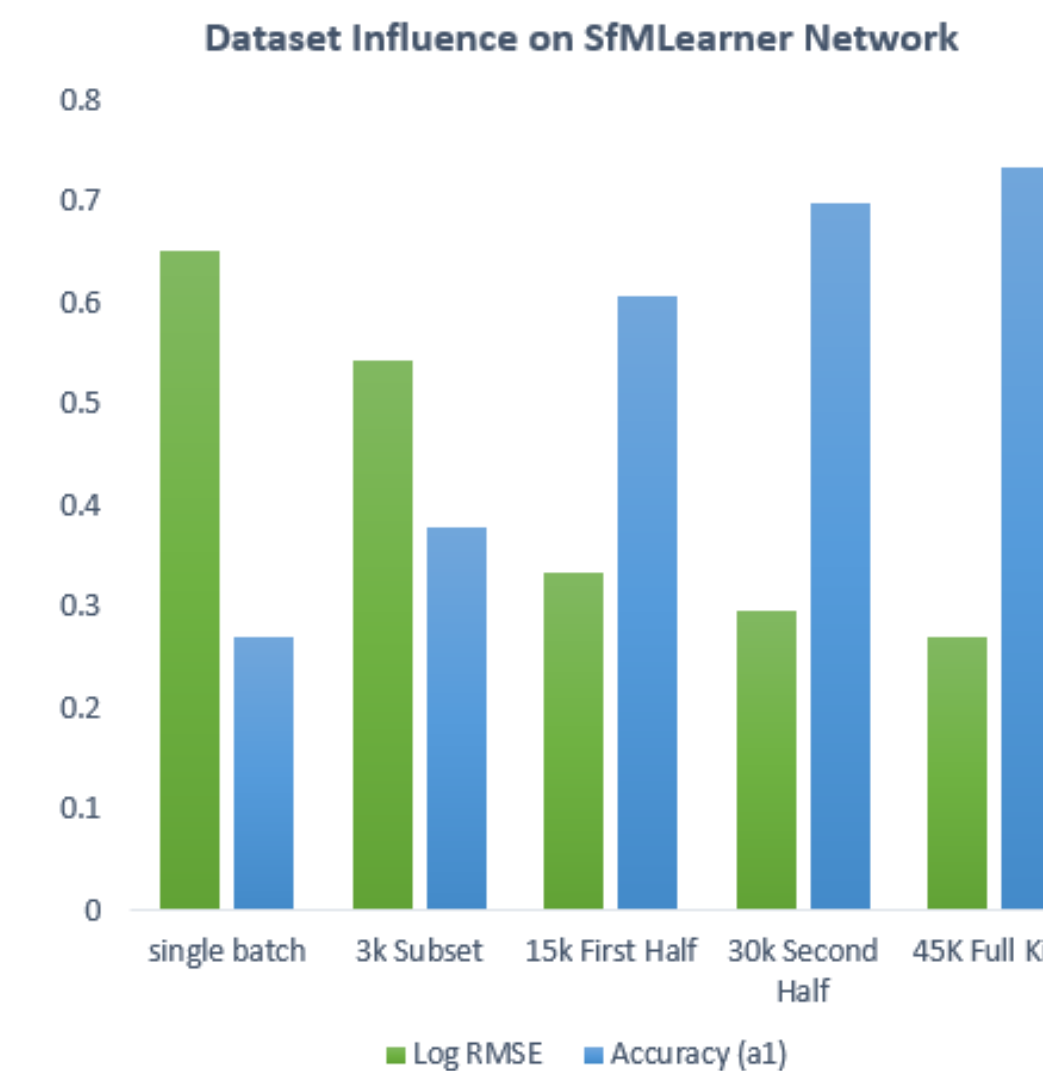
Importance of Dataset In Unsupervised Learning

Visualization



SfMLearner Architecture Tested on Kitti. Inspired by Naive Attempts to Train on A Small Dataset

Evaluation Results



Large Dataset Is Crucial For Unsupervised Methods.

Total Failure Of Training on My Own Static Camera Dataset



No Intermediate Depth Prediction and An Awful Point Cloud Result

- Reconstruction Loss Doesn't Depend on Depth Prediction If The Camera is Fixed.

$$p_s \sim K \hat{T}_{t \rightarrow s} \hat{D}_t(p_t) K^{-1} p_t$$

Conclusion & Next Step

- Discussed Methods for Depth Don't Generalize Well, Especially For Static Camera
- Scene Flow Back Project to Construction Loss Has Potential For Improvement
- Interactive Perception: Robots Can Provide Depth And Scene Flow Ground Truth